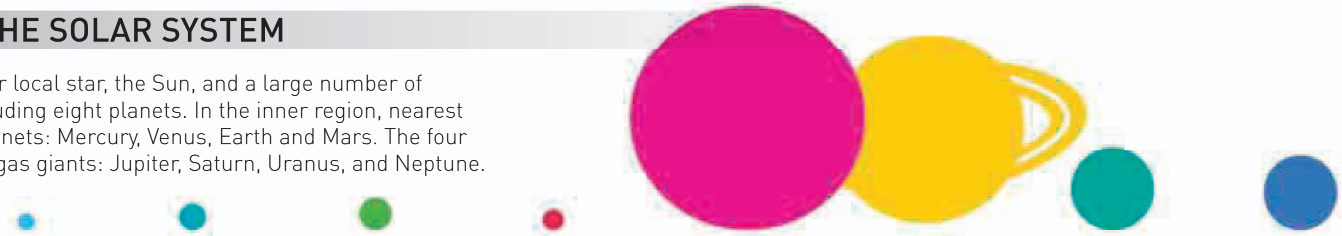
ASTRONOMY AND SPACE

THE PLANETS OF THE SOLAR SYSTEM

The Solar System consists of our local star, the Sun, and a large number of objects that orbit around it, including eight planets. In the inner region, nearest the Sun, there are four rocky planets: Mercury, Venus, Earth and Mars. The four outer planets are known as the gas giants: Jupiter, Saturn, Uranus, and Neptune.



PLANET	MERCURY	VENUS	EARTH	MARS	JUPITER	SATURN	URANUS	NEPTUNE
Distance from Sun millions of km (miles)	57.9 (36.0)	108.2 (67.2)	149.6 (93)	227.9 (141.5)	778.3 (483.3)	1,427 (886)	2,870 (1,782)	4,497 (2,774)
Diameter at equator km (miles)	4,879 (3,033)	12,104 (7,523)	12,756 (7,928)	6,786 (4,222)	142,984 (88,784)	120,536 (74,914)	51,118 (31,770)	49,528 (30,757)
Mass (Earth = 1)	0.06	0.82	1	0.11	317.83	95.16	14.54	17.15
Volume (Earth = 1)	0.056	0.86	1	0.15	1,319	744	67	57
Surface temperature °C (°F)	−180 to +430 (−356 to +800)	+480 (+896)	−70 to +55 (−158 to +133)	−120 to +25 (−248 to +77)	−150 (−238)	−180 (−292)	−214 (−353)	−220 (−364)
Surface gravity (Earth = 1)	0.38	0.91	1	0.38	2.64	0.92	0.79	1.12
Time to orbit Sun ("year")	87.9 days	224.7 days	365.3 days	687.0 days	11.9 years	29.5 years	84.0 years	164.8 years
Time to turn 360° ("day")	58.6 days	243.0 days	23.9 hours	24.6 hours	9.9 hours	10.7 hours	17.2 hours	16.1 hours
Orbital speed km/s (miles/s)	47.9 (29.7)	35.0 (21.7)	29.8 (18.5)	24.1 (15)	13.1 (8.1)	9.6 (6)	6.8 (4.2)	5.4 (3.4)
Number of observed moons	0	0	1	2	64	62	27	13

KEPLER'S LAWS OF PLANETARY MOTION

These laws, first formulated by 17th-century astronomer Johannes Kepler (1571–1630), show how the planets move around the Sun. The laws demonstrate that the planets travel in eliptical, not circular, orbits (see pp.100–101) and that the farther they are from the Sun, the slower the orbiting speed.

LAW	DESCRIPTION
First law	This law, sometimes called the law of ellipses, states that planets move around the Sun in a regular oval-shaped path, known as an ellipse, with the Sun at one focus. An ellipse has two focuses along its major axis. On any particular ellipse, the total distance from one focus to any point on the ellipse and back to the other focus is always the same.
Second law	Also known as the law of equal areas, this law describes how the speed of a planet changes as it orbits the Sun. A line drawn from the center of the Sun to the center of a planet will sweep out equal areas in equal intervals of time. Therefore, a planet moves faster when it is near the Sun and slower when it is farther away from it.
Third law	Also known as the law of harmonies, the third law describes the mathematical relationship between the distances of the planets from the Sun and their orbital periods. It states that the square of each planet's orbital period (the time it takes to travel one orbit of the Sun) is directly proportional to the cube of its average distance from the Sun. This law allows orbital period and distance to be calculated for each of the planets.

SPECTRAL CLASSIFICATION OF STARS

Light from a star can be split into a band of its component wavelengths called a spectrum. The positions of dark absorption lines and bright emission lines on this spectrum indicate the chemical components in the star's atmosphere. Based on their spectra, stars are divided into seven main classes—O, B, A, F, G, K, M.

TYPE	COLOUR	PROMINENT SPECTRAL LINES	AVERAGE TEMPERATURE	EXAMPLE OF STAR
O	Blue	He^+, He, H, O^{2+}, N^{2+}, C^{2+}, Si^{3+}	45,000°C (80,000°F)	Regor
B	Blueish white	He, H, C^+, O^+, N^+ Fe^{2+}, Mg^{2+}	30,000°C (55,000°F)	Rigel
A	White	H, ionized metals	12,000°C (22,000°F)	Sirius
F	Yellowish white	H, Ca^+, Ti^+, Fe^+	8,000°C (14,000°F)	Procyon
G	Yellow	H, Ca^+, Ti^+, Mg, H, some molecular bands	6,500°C (12,000°F)	The Sun
K	Orange	Ca^+, H, molecular bands	5,000°C (9,000°F)	Aldebaran
M	Red	TiO, Ca, molecular bands	3,500°C (6,500°F)	Betelgeuse